Smart India Hackathon 2026

SMART INDIA HACKATHON 2026

Team ID : ADE19D

Team Name : AgriBridge

Problem Statement ID : SIH26031

Problem statement Title : Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies.

Theme : Fitness & Sports

Category : Software

Evaluator Name : Anshul Bhanwalkar

Date of Evaluation : 03 - September - 2026

SIH

AGRIBRIDGE

AgriGrade AI

AI + IoT Fusion for Instant, Accurate & Traceable Quality Grading of Agricultural Produce

SMART INDIA HACKATHON 2026

THE PROBLEM

Agricultural grading today is **manual, inconsistent** and fails to detect **early-stage quality issues.**

Inconsistent & Subjective
Grading varies from person to person, leading to disputes and mistrust.

Early Quality Issues Go Unnoticed
Internal spoilage or defects are not visible but reduce shelf life and market value.

Size & Quality Not Accurate
Manual size estimation and quality checks are imprecise and non-standardized.

No Instant Digital Record
Paper-based records can be lost, tampered with or misused. No real-time traceability.

IMPACT:
Financial losses, farmer exploitation, quality inconsistencies & supply chain inefficiencies.

OUR SOLUTION

A Hybrid **AI + IoT System** that combines **Gas Sensing** and **Computer Vision** to detect both **Visible defects** and **Early quality issues** — delivering an **Instant, Accurate & Traceable Quality Grade** for any agricultural produce.

HOW IT WORKS

1. Sampling

Take a representative sample (1–2 kg) from the produce batch.

2. Dual Capture

Place sample in the smart sensing pod.

- IoT sensors capture gas emissions, temperature & humidity.
- Camera captures multiple images.

3. AI Analysis (Parallel)

Gas AI (IoT)
Detects early quality risks using gas signatures (Ethylene, VOCs, CO_2, and more) along with environmental data.
Output: Risk Score

Vision AI
Detects visible defects, blemishes, size distribution & foreign materials.
Output: Quality Score

4. Fusion & Decision
(AI Fusion Engine)

Combines both AI outputs with confidence scoring to generate a final Quality Grade aligned with standards.

Grade: A / B / C / D
Score: 87/100

5. Instant Report & Traceability

Grade A 87/100 → Digital Report (PDF) → QR Code → Cloud Sync & Secure Storage

- ✓ Timestamped
- ✓ Geo-tagged
- ✓ Tamper-proof
- ✓ Shareable instantly

Digital Quality Certificate

Batch ID	:	AG-2026-000782
Date	:	03 Sept 2026
Produce	:	Mixed Produce
Location	:	Procurement Center
Grade	:	A
Score	:	87/100

AGRIBRIDGE

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

Hybrid AI + IoT System for Onion Quality Grading

1. FIELD HARDWARE (Onion Sampling & IoT Device)

1 SENSING LAYER

Inside Sealed Sampling Chamber

- Ethane (C_2H_6) Sensor
- Methane (CH_4) Sensor
- CO_2 Sensor
- NH_3 Sensor
- pH / Moisture Probe
- DHT22 (Temp / Humidity)

ESP32 Microcontroller

BLE (Link)

2 EDGE PROCESSING LAYER

On-Device Processing & Power System

- Signal Conditioning
- Sensor Calibration
- Basic Thresholding
- Local Data Packaging

Power System

Rechargeable Li-ion Battery

Solar Trickle Charging

WiFi (Link)

3 CONNECTIVITY LAYER

Data Transmission & Local Buffer

BLE / WiFi (Device Link)

BLE / WiFi (Uplink)

Local Buffer (Offline Storage)

SD Card / Flash (Fallback)

2. CLOUD / APPLICATION

4 GATEWAY / PRESENTATION LAYER

Camera Capture Module

BLE Receiver Module

Report Viewer UI

Local Cache & Sync

5 API / GATEWAY LAYER

Secure Access & Routing

REST / GraphQL Gateway

Authentication & Authorization (JWT / Oauth 2.0)

Rate Limiting | Request Routing | API Management

6 APPLICATION SERVICES LAYER

Business Logic

Image Preprocessing Service
Denoise, Resize, Normalize

Vision Inference Service (CNN Model)
Defect Detection
Size / Grade Estimation

Gas & Chemical Signature Inference Service (ML)
Gas (C_2H_6, CH_4, CO_2, NH_3) + pH / Moisture
Spoilage Stage Classification

Fusion & Scoring Service
Combine Vision & Gas Outputs
Compute Final Score

Report Generation Service
Grade Report
Recommendations
Export / Share

Read / Write (REST / SDK)

Write (Ingestion) (MQTT / HTTP)

Read / Write (REST / SQL)

7 DATA LAYER

Storage & Persistence

Object Storage
(Images & Files)
e.g., S3 / MinIO / Azure Blob

Time-Series Database
(Sensor Logs)
e.g., InfluxDB / TimescaleDB

Relational Database
(Grading, Metedata, Users, Devices)
e.g., PostgreSQL

8 ML INFRASTRUCTURE LAYER

Model Lifecycle Management

Data Collection
(Images, Sensor Logs, Labels, Field Feedback)

Model Training Pipeline
(Offline / Scheduled Training)

Model Registry & Versioning
(Store, Version, Manage Models)

Model Deployment
(Serving Endpoints for Inference Services)

Retraining Trigger (Automated)

Field Feedback
(Grades, Corrections, User Feedback)

ICON LEGEND (Hardware)

Gas Sensor

- Sensor

Sensor

Probe

Temp / - Humidity Sensor

MCU / Chip

Battery

Solar Panel

WiFi

Bluetooth

Bluetooth

SD Card

PROTOCOL LEGEND

BL E – Bluetooth Low Energy
WiFi – Wireless LAN (802.11)
MQTT – Message Queuing Telemetry Transport
REST – Representational State Transfer
gPPC – Google Remote Procedure Call

Data Flow

Bidirectional Flow

Control / Feedback Flow

AGRIBRIDGE

FEASIBILITY & VIABILITY

Practical. Affordable. Scalable.

SMART INDIA HACKATHON 2026

1. TECHNICAL FEASIBILITY

Low-cost & proven hardware
ESP32, MQ gas sensors, pH, DHT22, camera.

Smartphone as gateway
No extra display or hardware required.

Practical AI models
Vision + gas signature models using lightweight ML.

Scalable infrastructure
Cloud-based with continuous retraining.

2. OPERATIONAL FEASIBILITY

Easy to use
Simple 4-step workflow; minimal training for inspectors.

Reliable connectivity
BLE / Wi-Fi; syncs when internet is available.

Built for field use
Portable, battery/solar powered, rugged for mandi conditions.

Instant digital reports
Timestamped, geo-tagged reports in seconds.

3. FINANCIAL VIABILITY

Low cost per center
₹ 8,000 – ₹ 10,000 per unit

Low operating cost
Minimal maintenance and long sensor life.

High ROI
Faster grading, fewer disputes and better decisions → savings.

Multiple revenue streams
Device sales, SaaS analytics, data insights.

4. DATA FEASIBILITY

Easy data collection
Gas + environmental data and images at mandis.

Ground truth available
Spoilage validated by cut sample inspection.

Data improves over time
More centers = more data = better models.

DATA SOURCES

- Procurement centers
- ICAR-DOGR / Research labs
- Pilot deployments
- User feedback & outcomes

5. LEGAL & REGULATORY FEASIBILITY

AGMARK aligned
Grading logic mapped with AGMARK standards.

Transparent & traceable
QR-based reports ensure auditability.

Data privacy & security
Farmer & transaction data stored securely and used responsibly.

6. SCALABILITY

Designed to scale
Works across multiple crops and procurement centers.

Cloud-based architecture
Supports thousands of devices and users.

Continuous improvement
Models get better with real-world data and feedback loop.

OVERALL VIABILITY

Technically feasible, operationally practical and financially viable. Deployable at scale with high impact on transparency, efficiency and farmer trust.

- Benefits Farmers
- Supports Govt. Procurement
- Improves Trust & Transparency
- Better Quality, Better Markets

AGRIBRIDGE

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

BENEFITS OF THE SOLUTION

GRADING ACCURACY	Reduces variation between inspectors, applies standardized quality criteria, and provides evidence behind grading decisions.
SPOILAGE INTELLIGENCE	Combines visible-defect detection with gas signatures, identifies potential early-stage deterioration, and helps distinguish borderline batches.
PROCUREMENT EFFICIENCY	Reduces dependence on manual visual inspection and inspector variability, supports faster batch-level decisions, and generates digital assessment records automatically.
TRACEABILITY	Maintains a digital record of each assessment, enables batch-wise quality history, and makes grading decisions easier to verify.
WASTE REDUCTION	Identifies batches with potential spoilage risk earlier, supports informed storage-duration decisions, and helps reduce avoidable post-harvest losses.

POTENTIAL IMPACT ON THE TARGET AUDIENCE

FARMERS	Receive an objective and evidence-based grade with digital records, reducing disputes and ensuring fair treatment during procurement.
PROCUREMENT CENTERS	Same standardized grading workflow across inspectors and locations ensures consistency, reliability and ease of adoption.
STORAGE & BUYERS	Gas sensor data reveals potential deterioration before visible defects appear, enabling better storage planning and informed buying decisions.
AGRICULTURAL SUPPLY CHAIN	Digital grading records create a valuable dataset for procurement planning, quality tracking, traceability and policy-level decision making.

OVERALL IMPACT

An intelligent, data-driven grading system that improves quality transparency, reduces losses, builds trust across the ecosystem, and strengthens the entire agricultural value chain.

- Fair & Transparent Grading
- Stronger Trust & Accountability
- Better Decisions & Planning
- Better Quality, Better Markets

AGRIBRIDGE.

RESEARCH & REFERENCES

Key research supporting our approach to AI + Gas Sensing based onion quality assessment and grading

SMART INDIA HACKATHON 2026

#	RESEARCH PAPER	KEY CONTRIBUTION	SOURCE
1	Quality assessment of onions using electronic nose — system development and detection	• Developed an electronic nose system for onion quality assessment. • Detected freshness and spoilage using volatile compounds. • Achieved good classification accuracy with sensor array.	ResearchGate
2	Detecting sour skin infected onions using a customized gas sensor array	• Designed a customized gas sensor array to detect sour skin. • Identified specific gas patterns associated with infected onions. • High detection accuracy in real storage conditions.	ScienceDirect
3	Onion sour skin detection using a gas sensor array and SVM	• Used gas sensor array with SVM model for sour skin detection. • Extracted features from sensor responses for classification. • Demonstrated effective and robust detection performance.	Springer
4	IoT-enabled electronic nose system for beef quality monitoring and spoilage detection	• IoT-based e-nose system for real-time quality and spoilage monitoring. • Cloud connectivity for data storage and remote access. • Early detection improved shelf-life and quality control.	PMC
5	Onion spoilage detection using PEDOT coated paper based sensor	• Developed PEDOT coated paper-based sensor for spoilage detection. • Sensitive to onion spoilage-related volatile compounds. • Low-cost, flexible and suitable for practical applications.	ResearchGate
6	Real-time visual inspection system for grading fruits using computer vision and deep learning techniques	• Uses deep learning (e.g., MobileNetV2) for vision-based grading. • High accuracy in automated fruit quality assessment. • Supports real-time, non-destructive inspection.	ScienceDirect

COMPARISON WITH EXISTING SYSTEMS

Feature	Our solution	Intello Labs	Agrograde	Manual grading
Vision defect detection	✓	✓	✓	✗
Gas or chemical spoilage sensing	✓	✗	✗	✗
Internal early rot detection	✓	✗	✗	✗
Field-deployable mobile app	✓	✓	✗	✗
Instant digital report	✓	✓	✓	✗
AGMARK-aligned grading	✓	✗	✗	✓
Low-cost hardware	✓	✓	✗	✓
Dispute-resistant record	✓	✓	✓	✗
Retraining from field data	✓	✓	✗	✗
Mandi-scale scalability	✓	✓	✓	✓

✓ Supported / Available   ✗ Not available / Limited

KEY TAKEAWAY

Research across gas sensors, AI/ML, computer vision and IoT demonstrates the potential to accurately detect early spoilage and automate onion quality grading.

SMART INDIA HACKATHON 2026

THANK YOU!

FOR YOUR TIME, SUPPORT & PARTICIPATION

Together, let's build a smarter, greener and more connected future.

SMARTER SOLUTIONS

Innovating for
real-world agri challenges.

CONNECTED FUTURE

AI + IoT + Blockchain
for seamless ecosystems.

TRUST & TRANSPARENCY

Secure, traceable
and reliable systems.

SUSTAINABLE IMPACT

Driving growth for
farmers, buyers & communities.

THANK YOU FOR BEING PART OF THIS JOURNEY!